

CHEMISTRY



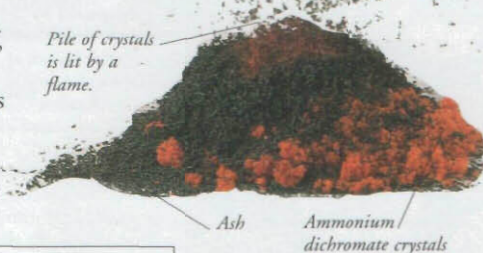
THERE IS MORE TO CHEMISTRY than messy experiments in laboratories – doctors use it to fight diseases, chefs use it to cook food, and farmers use it to increase the growth of their crops. Chemistry is the branch of science that studies the structure of different elements and compounds. It also investigates how they change and interact with each other during processes called chemical reactions.

Chemical change

When a pile of orange ammonium dichromate crystals is heated by a flame, a chemical reaction occurs. Heat, light, and gases are given off, and a mound of grey-green ash is left behind. The ash not only looks different from the crystals, but it also has a different chemical make-up – it has changed into the substance chromium oxide.

The reaction is so vigorous that a cloud of ash is hurled into the air.

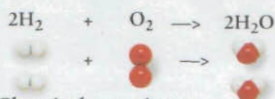
Pile of crystals is lit by a flame.



Ash
Ammonium dichromate crystals

Antoine and Marie Lavoisier

The French chemist Antoine Lavoisier (1743–94) showed that burning is a chemical reaction, that air is a mixture of gases, and that water is a compound of hydrogen and oxygen. His wife, Marie (1758–1836), translated and illustrated many of his scientific works.



Chemical equations

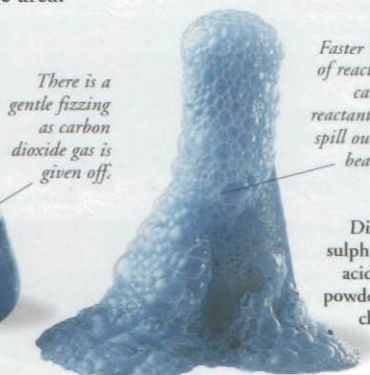
Scientists write equations to describe what happens during reactions. The equation above shows how hydrogen (H_2) and oxygen (O_2) react in the ratio of 2 to 1 to make water (H_2O).

Rates of reaction

Reactions can be speeded up by making the reacting particles come into contact with each other more often. One way of doing this is by increasing a reactant's surface area. Sulphuric acid reacts more rapidly with powdered chalk than with chalk pieces, because the powder has a greater surface area.



Dilute sulphuric acid on chalk pieces



Faster rate of reaction causes reactants to spill out of beaker.

Dilute sulphuric acid on powdered chalk

Catalysts

Compounds called catalysts speed up a chemical reaction by helping substances react together. The catalysts are left unchanged by the reaction. Many cars are fitted with a catalytic converter to remove polluting gases from engine fumes. The converter forces the gases into close contact with catalysts. The catalysts make the gases react rapidly with each other, producing less harmful gases that escape out of the exhaust.

Chemical reactions

During a chemical reaction, substances called reactants break apart and new substances called products form. Energy is taken in to break the bonds between the reactants' atoms. As the atoms link up again in different combinations to make the products, new bonds form and energy is given out.



Exothermic reactions

In an exothermic reaction, such as burning, more energy is given out than is taken in from the surroundings.



Endothermic reactions

Most of the reactions that occur in cooking are endothermic, which means that more energy is taken in than is given out.

Oxidation and reduction

When iron rusts, a reaction occurs between the iron and oxygen in the air. The iron gains oxygen, and an orange-brown compound called iron oxide forms. A reaction in which a substance gains oxygen is called oxidation. When oxidation occurs, there is a simultaneous reaction called reduction, in which a substance loses oxygen. When iron oxidizes, the air is reduced as it loses oxygen to the iron.

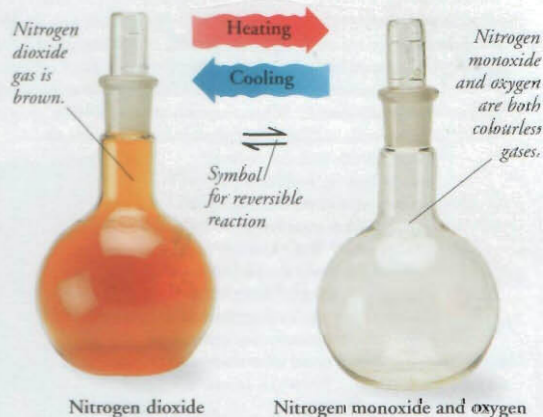


Coating of iron oxide forms on the metal.

Rusting iron

Reversible reactions

Many chemical reactions permanently change the reactants, but reversible reactions can go both forwards and backwards. For example, when nitrogen dioxide is heated, it breaks down into nitrogen monoxide and oxygen. Cooling this mixture makes the two gases react to form nitrogen dioxide again.



Chemical industry

The chemical industry is one of the world's largest and most important industries. It involves taking raw materials – such as air, oil, water, coal, metal ores, limestone, and plants – and using chemical reactions to change them into useful products. These products include food, clothing, medicine, pesticides and fertilizers, paints and dyes, soaps and detergents, plastics, and glassware.

